



Memo version 2.0

# BESIII Analysis Memo

BAM-xxx

BESIII Analysis Memo (March 9, 2026)

## Cross section measurement of process $e^+e^- \rightarrow \pi^+\pi^-\pi^0$ from threshold to 2.0 GeV via ISR technique

Guang-Yan Xiao <sup>a,c1</sup> Ya-Qian Wang <sup>b</sup> Lei Zhang <sup>a</sup> Shen-Jian Chen <sup>a</sup> Chang-Zheng Yuan <sup>c</sup>

<sup>a</sup>Nanjing University

<sup>b</sup>Hebei University

<sup>c</sup>Institute of High Energy Physics, CAS

## Referee Committee

XXX (Chair)<sup>d</sup>, XXX<sup>e</sup>, XXX<sup>f</sup>

<sup>d</sup>XXX University

<sup>e</sup>XXX University

<sup>f</sup>XXX University

## Pubcomm Editors

Editor A<sup>g</sup>, Editor B<sup>h</sup>

<sup>g</sup>Uni. G

<sup>h</sup>Uni. H

HN : <http://hnbcs3.ihep.ac.cn/HyperNews/get/paper835.html>

## Abstract

XXX

<sup>1</sup>E-mail: xiaogy@smail.nju.edu.cn

20 **List of changes**

21 For an easy reading of the document, major changes in various versions with respect to the previous  
22 one are listed below.

23 **Version 1.0**

- 24 • First version of memo.

|    |                                                                     |           |
|----|---------------------------------------------------------------------|-----------|
| 25 | <b>Contents</b>                                                     |           |
| 26 | <b>1 Introduction</b>                                               | <b>3</b>  |
| 27 | <b>2 BEPCII and BESIII detector</b>                                 | <b>4</b>  |
| 28 | <b>3 Data and MC samples</b>                                        | <b>5</b>  |
| 29 | <b>4 Event selections</b>                                           | <b>7</b>  |
| 30 | <b>5 Background Study</b>                                           | <b>9</b>  |
| 31 | 5.1 Estimate of $e^+e^- \rightarrow \pi^+\pi^-\pi^0\pi^0$ . . . . . | 13        |
| 32 | 5.2 Estimate of Other Backgrounds and Signal Yields . . . . .       | 16        |
| 33 | <b>6 Unfolding</b>                                                  | <b>19</b> |
| 34 | 6.1 The IDS unfolding method . . . . .                              | 20        |
| 35 | 6.2 Parameter optimization and figure of merit . . . . .            | 21        |
| 36 | <b>7 Correction of response matrix, and unfolding of data</b>       | <b>24</b> |
| 37 | 7.1 Tracking and PID efficiencies . . . . .                         | 24        |
| 38 | 7.2 The $\pi^0$ reconstruction efficiency . . . . .                 | 24        |
| 39 | 7.3 The ISR photon detection efficiency . . . . .                   | 27        |
| 40 | 7.4 PHOKHARA Extra Photon Fraction . . . . .                        | 31        |
| 41 | 7.4.1 Results for Large Angle (LA) NLO photons . . . . .            | 32        |
| 42 | 7.4.2 Results for Small Angle (SA) NLO photons . . . . .            | 32        |
| 43 | <b>8 Systematic uncertainties</b>                                   | <b>33</b> |
| 44 | <b>9 Summary</b>                                                    | <b>34</b> |

**1 Introduction**

to fill

## 2 BEPCII and BESIII detector

The Beijing Electron-Positron Collider II (BEPCII) —a double-ring multi-bunch collider—is operating in the  $\tau$ -charm energy region (2 to 4.9 GeV) with the design luminosity of  $1 \times 10^{33} \text{ cm}^{-2}\text{s}^{-1}$  [1]. The Beijing Spectrometer III (BESIII) operating at BEPCII is a high-precision, general-purpose detector that records symmetric  $e^+e^-$  collisions provided by the BEPCII storage ring. BESIII is approximately cylindrically symmetric with 93% coverage of the solid angle around the  $e^+e^-$  interaction point (IP). The components of the apparatus ordered by distance from the IP, are a 43-layer small-cell main drift chamber (MDC), a time-of-flight (TOF) system based on plastic scintillators with two layers in the barrel region and one layer in the end-cap region (the end-cap TOF system has been updated with the multi-gap resistive plate chamber in 2014 [2]), a 6240-cell CsI(Tl) crystal electromagnetic calorimeter (EMC), a superconducting solenoid magnet providing a 1.0 T magnetic field aligned with the beam axis, and resistive plate muon-counter layers interleaved with steel. The momentum resolution for charged tracks in the MDC is 0.5% for transverse momenta with 1 GeV/c. The energy resolution in the EMC is 2.5% in the barrel region and 5.0% in the end-cap region for 1 GeV photons. For charged tracks, particle identification (PID) for charged tracks combines measurements of the energy deposit  $dE/dx$  in MDC and flight time in TOF and forms likelihoods  $\mathcal{L}(h)(h = K, \pi)$  for a hadron  $h$  hypothesis. More details about the BESIII detector are provided in Ref [3].

Table 1: Beam energies and integrated luminosities of the data samples used in this work, quoted from Ref. [4, 5].

| $E_{\text{beam}}(\text{GeV})$ | Integrated luminosity ( $\text{pb}^{-1}$ ) | Data taking time               | RunNo                                         |
|-------------------------------|--------------------------------------------|--------------------------------|-----------------------------------------------|
| 3.773                         | $2931.8 \pm 0.2 \pm 13.8$ [4]              | 2010(round03)<br>2011(round04) | 11414 – 13988, 14395 – 14604<br>20448 – 23454 |
| 3.773                         | $4995.0 \pm 19.0$ [5]                      | 2022(round15)                  | 70522 – 73929                                 |
| 3.773                         | $8157.0 \pm 31.0$ [5]                      | 2023(round16)                  | 74031 – 78536                                 |
| 3.773                         | $4191.0 \pm 16.0$ [5]                      | 2024(round17)                  | 78615 – 81094                                 |

Table 2: Processes and luminosity scales of the inclusive MC samples used in this analysis. Separate samples are available for each data-taking round. Most processes use samples with an integrated luminosity equivalent to ten times that of the data.

| Process                                                         | Luminosity scale |
|-----------------------------------------------------------------|------------------|
| $e^+e^- \rightarrow \psi(3770) \rightarrow D^0\bar{D}^0$        | $10 \times$      |
| $e^+e^- \rightarrow \psi(3770) \rightarrow D^+D^-$              | $10 \times$      |
| $e^+e^- \rightarrow \tau^+\tau^-$                               | $10 \times$      |
| $e^+e^- \rightarrow \psi(3770) \rightarrow \text{non-}D\bar{D}$ | $10 \times$      |
| $e^+e^- \rightarrow q\bar{q}$                                   | $10 \times$      |
| $e^+e^- \rightarrow \gamma_{\text{ISR}}\psi(2S)$                | $10 \times$      |
| $e^+e^- \rightarrow \gamma_{\text{ISR}}J/\psi$                  | $10 \times$      |
| $e^+e^- \rightarrow e^+e^-$                                     | $0.25 \times$    |
| $e^+e^- \rightarrow \gamma\gamma$                               | $3 \times$       |
| $e^+e^- \rightarrow \mu^+\mu^-$                                 | $15 \times$      |

### 3 Data and MC samples

We use an  $e^+e^-$  collision sample corresponding to an integrated luminosity of  $20 \text{ fb}^{-1}$ , collected by BESIII detector at center-of-mass energy of 3.773 GeV, under BOSS-7.1.2 version. The details are listed in Table 1, quoted from Ref. [4, 5].

Simulated Monte Carlo (MC) samples are employed to study backgrounds, optimize event selection, and estimate detection efficiencies. Two categories of MC samples are utilized in this analysis:

- **The inclusive MC samples** comprehensively model known physics processes at  $\sqrt{s} = 3.773 \text{ GeV}$ , including  $D\bar{D}$  production,  $\tau^+\tau^-$  pair production, initial-state-radiation (ISR) production of vector charmonium states ( $J/\psi$  and  $\psi(2S)$ ), QED processes, and continuum light hadron production. The charm working group provides separate samples for the different data-taking rounds(round03-04, round15, round16, and round17) [7]. In this analysis, we use inclusive MC samples with an integrated luminosity equivalent to ten times that of the data for most processes, except for the QED processes  $e^+e^- \rightarrow e^+e^-$ ,  $\gamma\gamma$ , and  $\mu^+\mu^-$ , where we use the samples with the scaling factors provided by the charm working group. The processes and the corresponding luminosity scales of

the samples used in this analysis are summarized in Table 2.

- **Exclusive signal and background MC samples** generated specifically for this analysis using the PHOKHARA event generator [8, 9, 10] (version 9.1). These samples are produced independently for each data-taking round (round03, round04, round15, round16, round17) to account for variations in detector conditions. Sufficient statistics are generated for each process to ensure reliable efficiency determination and background studies. The generated processes include:

$$- e^+e^- \rightarrow \pi^+\pi^-\pi^0\gamma_{\text{ISR}} \text{ (signal process)}$$

$$- e^+e^- \rightarrow \pi^+\pi^-\pi^0\pi^0$$

$$- e^+e^- \rightarrow \pi^+\pi^-\pi^0\pi^0\gamma_{\text{ISR}}$$

These exclusive samples are used for signal efficiency studies, background characterization, and validation of the analysis methodology.

## 4 Event selections

The process under investigation is  $e^+e^- \rightarrow \pi^+\pi^-\pi^0\gamma_{\text{ISR}}$ , whose final states include two charged tracks, one neutral  $\pi^0$ , and one ISR photon. The following selection criteria is applied to all the data samples. Here the figures are shown by taking  $\sqrt{s} = 3.773\text{GeV}$  round16 data as an example. The distributions related to the event selection of other data sets are shown in Appendix. XXX. Situations are similar to each other.

- Good charged track: A charged track is taken as a good MDC track if it originates from the beam vertex and lies within the MDC polar angle region:
  - $|\cos\theta| < 0.93$ , where  $\theta$  is the polar angle with respect to the beam axis.
  - $|\Delta r| < 1\text{ cm}$ ,  $|\Delta z| < 10\text{ cm}$ , where  $|\Delta r|$  and  $|\Delta z|$  is distance of closest approach to the interaction point (IP) perpendicular to the beam axis and along the beam axis, respectively.
  - the ratio of the calorimeter-deposited energy to the track momentum  $E_{\text{EMC}}/p$  is required to be less than 0.8 for both of the charged tracks.
  - no PID requirements for  $\pi^+$ ,  $\pi^-$ .
- $\gamma$  selection:
  - $E > 0.025\text{ GeV}$  (barrel:  $|\cos\theta| < 0.8$ ),  $E > 0.050\text{ GeV}$  (endcap:  $0.86 < |\cos\theta| < 0.92$ ).
  - $0 \leq t_{\text{TDC}} \leq 14(\times 50\text{ns})$  to suppress fake photons due to electronic noise or beam background.
  - the closest angle with all charged tracks  $\theta > 10^\circ$  to suppress background from radiation of charged tracks.
- $\pi^0 \rightarrow \gamma\gamma$  selection:
  - two photon invariant mass  $M_{\gamma\gamma}$  within  $(0.115, 0.150)\text{ GeV}/c^2$ .
  - To improve the momentum resolution, 1C kinematic fit is performed. Here, the 1C means  $\pi^0$  mass constraint, and the updated 4-momentum will be used in the further analysis.  $\chi_{\text{KF}}^2 < 10$

Candidate events are selected by requiring exactly two good charged tracks, accompanied by at least one reconstructed  $\pi^0$  and a minimum of three good photon showers. This ensures the presence of at least one  $\pi^0$  and one isolated photon from Initial-State Radiation ( $\gamma_{\text{ISR}}$ ). A four-constraint (4C) kinematic fit is then applied to the event, and the candidate with the smallest  $\chi_{4\text{C}}^2$  value is chosen for further analysis.

After identifying an ISR photon from the isolated photon showers, the invariant mass of the remaining  $e^+e^- \rightarrow \pi^+\pi^-\pi^0$  system is reconstructed. As expected for ISR processes, the resulting center-of-mass energy  $\sqrt{s'}$  of the  $\pi^+\pi^-\pi^0$  system shows a continuous spectrum, characteristic of the radiative return mechanism. This continuum behavior is illustrated in Fig. 1 across the  $\omega(782)$ ,  $\phi(1020)$ , and  $\omega'/\omega''$  mass regions.



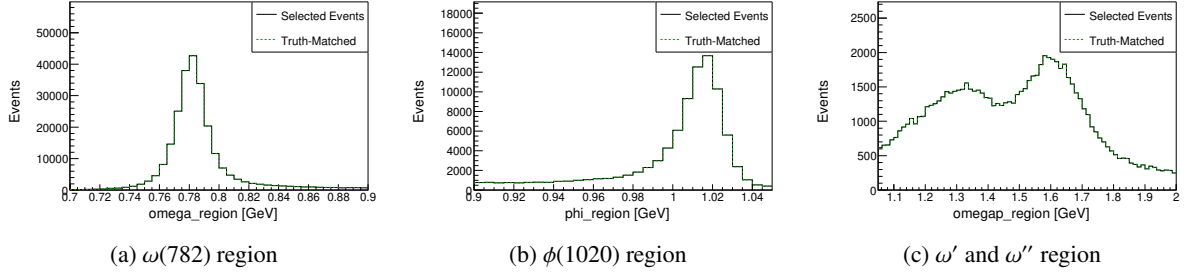


Figure 1: Invariant mass spectra of MC in the (a)  $\omega(782)$ , (b)  $\phi(1020)$ , and (c)  $\omega'/\omega''$  mass regions. The distributions correspond to  $M(pK_S^0)$ ,  $M(p\pi^0)$ , and  $M(K_S^0\pi^0)$ , respectively. The "Truth-Matched" distribution (green dashed line) corresponds to reconstructed events where the momentum directions of the  $\pi^0$  and  $\gamma_{\text{ISR}}$  successfully match their truth-level counterparts. A loose requirement of  $\chi_{\text{KF}}^2 < 100$  is applied to the kinematic fit. The continuous spectrum of  $\sqrt{s}$  for the  $\pi^+\pi^-\pi^0$  system demonstrates the radiative return mechanism via ISR.

## 5 Background Study

In the study of background processes, the main backgrounds are categorized into  $e^+e^- \rightarrow \pi^+\pi^-\pi^0\pi^0$ ,  $e^+e^- \rightarrow \pi^+\pi^-\pi^0\pi^0\gamma_{\text{ISR}}$ , QED processes, and others. The dominant contributions arise from processes with additional  $\pi^0$ , particularly  $e^+e^- \rightarrow \pi^+\pi^-\pi^0\pi^0$ , while  $e^+e^- \rightarrow \pi^+\pi^-\pi^0\pi^0\gamma_{\text{ISR}}$  is not dominant but shares the same underlying mechanism as a background to our signal. In the inclusive analysis, the cross sections used for these two processes were not correctly configured.

To verify that these processes constitute the primary backgrounds, the  $\mathcal{L} \times \sigma_{\text{hadron}}$  method is employed, where  $\mathcal{L}$  is the integrated luminosity and  $\sigma_{\text{hadron}}$  is the hadronic cross section. The luminosity values for each data-taking period are listed in Table 1. The background yield is estimated as a function of the  $\pi^+\pi^-\pi^0$  invariant mass ( $m$ ) using the effective luminosity spectrum, which is related to the mass spectrum via:

$$\frac{dL_{\text{eff}}}{dm} = \frac{2m}{s} \cdot F(s, m) \cdot \mathcal{L}_{\text{int}}, \quad (1)$$

where  $s = 4E^2$ ,  $E$  is the beam energy,  $\mathcal{L}_{\text{int}}$  is the total integrated luminosity, and  $F(s, m)$  is the probability function for initial-state photon emission. This function can be written as [11]:

$$\beta = \frac{2\alpha}{\pi} \left[ \ln\left(\frac{s}{m_e^2}\right) - 1 \right], \quad (2)$$

$$x = 1 - \frac{m^2}{s}, \quad (3)$$

$$\begin{aligned} \frac{dL_{\text{eff}}}{dm} = \frac{2m}{s} \cdot & \left\{ \beta x^{\beta-1} \left[ 1 + \frac{\alpha}{\pi} \left( \frac{\pi^2}{3} - \frac{1}{2} \right) + \frac{3}{4}\beta \right. \right. \\ & \left. \left. + \beta^2 \left( \frac{37}{96} - \frac{\pi^2}{12} - \frac{1}{72} \ln \frac{s}{m_e^2} \right) \right] \right. \\ & \left. - \beta \left( 1 - \frac{x}{2} \right) \ln \frac{1}{x} - \frac{1 + 3(1-x)^2}{x} \ln(1-x) - 6 + x \right\} \cdot \mathcal{L}_{\text{int}}, \quad (4) \end{aligned}$$

where  $\alpha = 1/137$  is the fine-structure constant,  $m_e = 0.5110034 \times 10^{-3} \text{ GeV}/c^2$  is the electron mass, and  $\sqrt{s} = 3.773 \text{ GeV}$  is the center-of-mass energy of the primary  $e^+e^-$  system.

The effective luminosity spectrum calculated using this formula for round16 is shown in Fig. 2. Similar calculations were performed for all data-taking rounds.

Based on this method, rough yield estimates for the two processes with additional  $\pi^0$ ,  $e^+e^- \rightarrow \pi^+\pi^-\pi^0\pi^0$  (non-ISR) and  $e^+e^- \rightarrow \pi^+\pi^-\pi^0\pi^0\gamma_{\text{ISR}}$  (ISR), were calculated for the four data-taking periods (round03/04, round15, round16, round17). The results are summarized in Table 3. For the ISR channel, the estimation relies on the  $e^+e^- \rightarrow \pi^+\pi^-\pi^0\pi^0$  cross section measured by the BaBar collaboration [12].

It is important to note that the values listed in the table are initial yield estimates without any selection efficiency corrections; the actual background contribution to the signal region is obtained by multiplying these yields by the corresponding selection efficiency relevant to the specific analysis cuts. These estimates serve only to confirm that these processes constitute the main sources of background and to provide guidance for optimizing subsequent selection criteria.

The final background yield is then obtained by multiplying the effective luminosity spectrum by

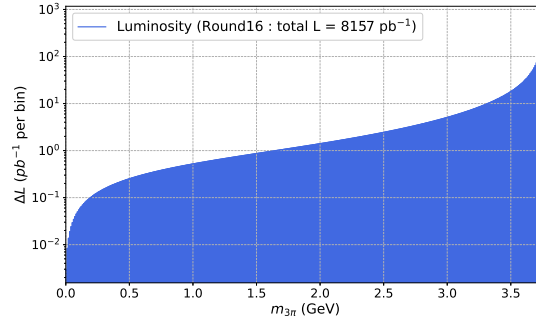


Figure 2: Effective luminosity spectrum as a function of the  $\pi^+\pi^-\pi^0$  invariant mass ( $m_{3\pi}$ ) calculated for round16 with total integrated luminosity  $\mathcal{L}_{\text{int}} = 8157 \text{ pb}^{-1}$ . The spectrum is shown with logarithmic scale on the vertical axis.

Table 3: Rough estimates of the expected yields for the two processes with additional  $\pi^0$  before applying any selection efficiency. The integrated luminosities are taken from Table 1. The cross section for  $e^+e^- \rightarrow \pi^+\pi^-\pi^0\pi^0$  is taken as  $\sigma = 0.22 \text{ nb}$  at  $\sqrt{s} = 3.773 \text{ GeV}$ , while the energy-dependent cross section for the ISR process is derived from the measured  $e^+e^- \rightarrow \pi^+\pi^-\pi^0\pi^0$  cross section [12]. The final background contribution is obtained by multiplying these yields by the corresponding selection efficiency.

| Data sample (Run period) | $N(e^+e^- \rightarrow \pi^+\pi^-\pi^0\pi^0)$ | $N(e^+e^- \rightarrow \pi^+\pi^-\pi^0\pi^0\gamma_{\text{ISR}})$ |
|--------------------------|----------------------------------------------|-----------------------------------------------------------------|
| round03-04               | 511940                                       | 1142096                                                         |
| round15                  | 872209                                       | 1945825                                                         |
| round16                  | 1424346                                      | 3177597                                                         |
| round17                  | 731817                                       | 1632623                                                         |

the cross section and a selection-efficiency factor that varies dynamically with the analysis cuts. This method is used solely to confirm that  $e^+e^- \rightarrow \pi^+\pi^-\pi^0\pi^0$  and  $e^+e^- \rightarrow \pi^+\pi^-\pi^0\pi^0\gamma_{\text{ISR}}$  are the dominant backgrounds and to optimize the selection criteria. It is not used to derive the final background estimates; precise yields for these two components will be determined using data-driven methods.

Based on the rough background estimate from  $\mathcal{L} \times \sigma_{\text{hadron}}$ , the signal yield was approximated by subtracting the background from the data, allowing for a preliminary significance calculation. Using this approach, a figure of merit (FOM) defined as  $S/\sqrt{S+B}$  was constructed to evaluate the signal significance. To optimize the event selection, a scan over the kinematic fit  $\chi^2$  was performed, with the result shown in Fig. 3. The Round 16 optimization yields an optimal cut of  $\chi_{\text{KF}}^2 < 50$ .

Based on the optimal  $\chi_{\text{KF}}^2$  selection, the invariant mass spectrum of the  $3\pi$  system ( $M_{3\pi}$ ) can be divided into three distinct regions for further study: the  $\omega(782)$  region, the  $\phi(1020)$  region, and a higher-mass region associated with possible  $\omega'/\omega''$  excitations. In the high-mass  $\omega'/\omega''$  region, a significant background component from events containing additional  $\pi^0$  mesons was observed. To quantify this, the number of  $\pi^0$  candidates per event was counted, where any  $\pi^0$  candidate satisfying a one-constraint kinematic fit  $\chi^2 < 200$  was accepted. The distribution of this  $\pi^0$  multiplicity is shown in Fig. 4. To suppress this specific background, a requirement of exactly one  $\pi^0$  candidate per event was applied

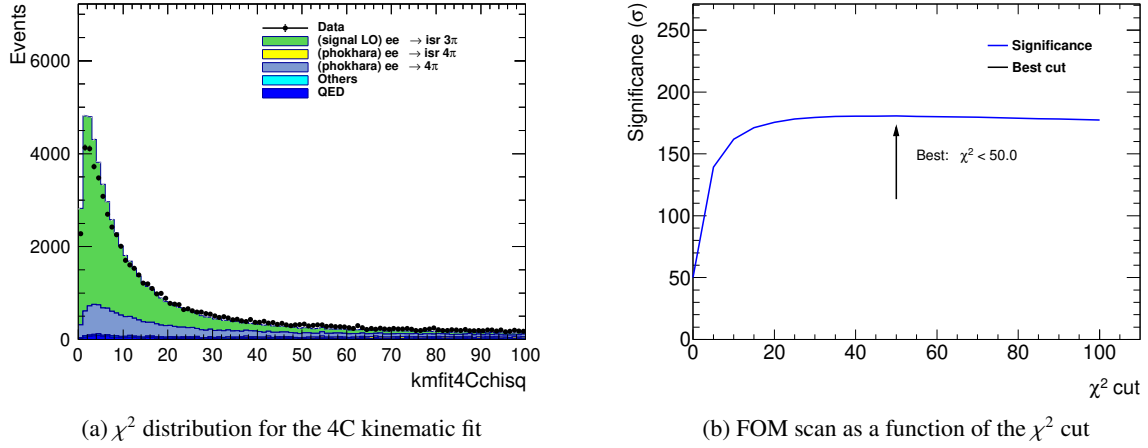


Figure 3: (a) Distribution of the kinematic fit  $\chi^2$  for the four-constraint (4C) fit using the Round 16 data sample. (b) The corresponding figure of merit ( $S/\sqrt{S+B}$ ) as a function of the applied  $\chi^2$  cut value for the Round 16 data.

164 exclusively in the  $\omega'/\omega''$  mass region. No such restriction was applied to the  $\omega(782)$  and  $\phi(1020)$  signal  
 165 regions. The following studies and distributions are based on the Round 16 data sample.

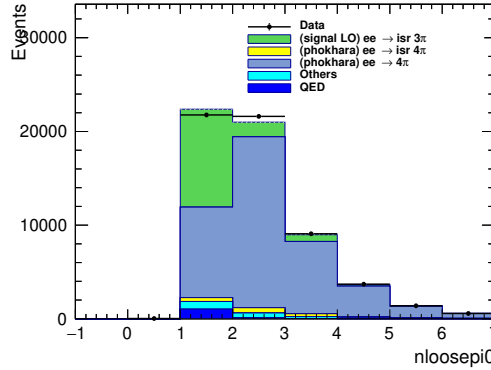


Figure 4: Distribution of the number of  $\pi^0$  candidates per event in the  $\omega'/\omega''$  mass region using the Round 16 data. A  $\pi^0$  candidate is counted if its 1C kinematic fit  $\chi^2 < 200$ .

166 The resulting  $M_{3\pi}$  invariant mass spectra, after applying all selection criteria including the region-  
 167 specific  $\pi^0$  multiplicity cut, are presented separately for the three mass regions in Fig. 5. The  $e^+e^- \rightarrow$   
 168  $\pi^+\pi^-\pi^0\pi^0\gamma_{\text{ISR}}$  contribution is negligible and will not be treated as a separate background in the subse-  
 169 quent fitting procedure; only the non-ISR  $e^+e^- \rightarrow \pi^+\pi^-\pi^0\pi^0$  process will be explicitly estimated using  
 170 data-driven methods and constrained in the fit.



## 5.1 Estimate of $e^+e^- \rightarrow \pi^+\pi^-\pi^0\pi^0$

In estimating the  $e^+e^- \rightarrow \pi^+\pi^-\pi^0\pi^0$  background, the following strategy was adopted:

It is assumed that the total yield of  $e^+e^- \rightarrow \pi^+\pi^-\pi^0\pi^0$  events produced in the data is a fixed quantity. This process enters our signal region primarily when a photon from one of the  $\pi^0$  decays is misidentified as an initial-state radiation photon ( $\gamma_{\text{ISR}}$ ). Consequently, two distinct selection efficiencies can be calculated:  $\epsilon_{\text{sig}}$  (the efficiency for  $e^+e^- \rightarrow \pi^+\pi^-\pi^0\pi^0$  events to be selected as signal candidates) and  $\epsilon_{\text{bkg}}$  (the efficiency for the same process to enter the  $3\pi$  signal region as background).

When  $e^+e^- \rightarrow \pi^+\pi^-\pi^0\pi^0$  events are selected as signal, the same analysis procedure as for the signal is applied to construct the  $3\pi$  invariant mass spectrum, thereby obtaining the distribution of this process in data. Using the ratio of these two efficiencies,  $R = \epsilon_{\text{bkg}}/\epsilon_{\text{sig}}$ , the observed  $e^+e^- \rightarrow \pi^+\pi^-\pi^0\pi^0$  signal distribution can be converted into an estimate of its background distribution in the signal region. Specifically:

$$N_{\text{bkg}} = R \times N_{\text{sig}}^{\text{obs}} = \frac{\epsilon_{\text{bkg}}}{\epsilon_{\text{sig}}} \times N_{\text{sig}}^{\text{obs}}, \quad (5)$$

where  $N_{\text{sig}}^{\text{obs}}$  is the number of observed  $e^+e^- \rightarrow \pi^+\pi^-\pi^0\pi^0$  signal events in data, and  $N_{\text{bkg}}$  is the estimated number of background events from this process in the signal region.

This method provides a data-driven conversion from the observed signal distribution to the background distribution through efficiency correction.

When treating  $e^+e^- \rightarrow \pi^+\pi^-\pi^0\pi^0$  as the signal, the selection criteria are designed such that the same event sample also satisfies a loose selection for  $e^+e^- \rightarrow \gamma_{\text{ISR}}\pi^+\pi^-\pi^0$ . This ensures that the invariant mass of the three-pion system is automatically defined within the latter process.

The combined selection proceeds as follows: events are required to meet the criteria for both processes. The first step applies the full  $e^+e^- \rightarrow \pi^+\pi^-\pi^0\pi^0$  selection:

### • $\pi^+\pi^-\pi^0\pi^0$ Candidate Selection:

- Exactly two good charged tracks.
- $n\text{Pi0} \geq 2$  &  $n\text{GoodShowers} \geq 4$  (ensuring at least two  $\pi^0$ s).
- Application of a total 4-momentum constraint to the initial  $e^+e^-$  collision (4C kinematic fit).
- The candidate with the minimum  $\chi_{4C}^2$  is chosen.
- A requirement of  $\chi_{\text{KF}}^2 < 10$  on the kinematic fit.

Subsequently, within the same event, a candidate for  $e^+e^- \rightarrow \gamma_{\text{ISR}}\pi^+\pi^-\pi^0$  is formed by reinterpreting one of the  $\pi^0$ s as the  $\gamma_{\text{ISR}}$ :

### • $\gamma_{\text{ISR}}\pi^+\pi^-\pi^0$ Candidate Selection:

- The same two good charged tracks.
- $n\text{Pi0} \geq 1$  &  $n\text{GoodShowers} \geq 3$  (ensuring at least one  $\pi^0$  and one photon assigned as  $\gamma_{\text{ISR}}$ ).
- Application of the same total 4-momentum constraint to the initial  $e^+e^-$  collision (4C kinematic fit).
- The candidate with the minimum  $\chi_{4C}^2$  is chosen.

– No additional kinematic fit  $\chi^2$  requirement is applied at this stage.

This dual-tag strategy guarantees that the reconstructed  $3\pi$  system from the  $e^+e^- \rightarrow \gamma_{\text{ISR}}\pi^+\pi^-\pi^0$  hypothesis is kinematically well-defined, and its invariant mass can be reliably calculated for the subsequent analysis.

The selection described above results in a highly pure sample of  $e^+e^- \rightarrow \pi^+\pi^-\pi^0\pi^0$  events. The invariant mass spectrum of the corresponding three-pion system from this pure sample, which serves as the input template for background estimation, is shown in Fig. 6.

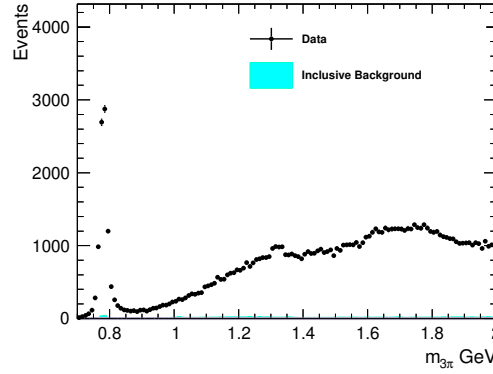


Figure 6: The  $3\pi$  invariant mass spectrum from selected  $e^+e^- \rightarrow \pi^+\pi^-\pi^0\pi^0$  events in the Round 16 dataset. This distribution, obtained from data after applying the dual-tag selection criteria for  $e^+e^- \rightarrow \pi^+\pi^-\pi^0\pi^0$ , represents the pure signal sample used as the template for background estimation.

The observed distribution in the data, after applying the  $e^+e^- \rightarrow \pi^+\pi^-\pi^0\pi^0$  selection criteria, will be scaled by the efficiency ratio  $R = \epsilon_{\text{bkg}}/\epsilon_{\text{sig}}$  to estimate the final background contribution of this process in the signal region of the measurement channel. As established and shown in Fig. 6, the background contamination within this selected sample is negligible. Therefore, we subtract the inclusive background histogram from the data histogram in a symbolic manner to obtain the pure  $e^+e^- \rightarrow \pi^+\pi^-\pi^0\pi^0$  event yield. The resulting background-subtracted signal distributions in the three  $3\pi$  invariant mass regions are presented in Fig. 7. All other backgrounds will be subtracted directly.

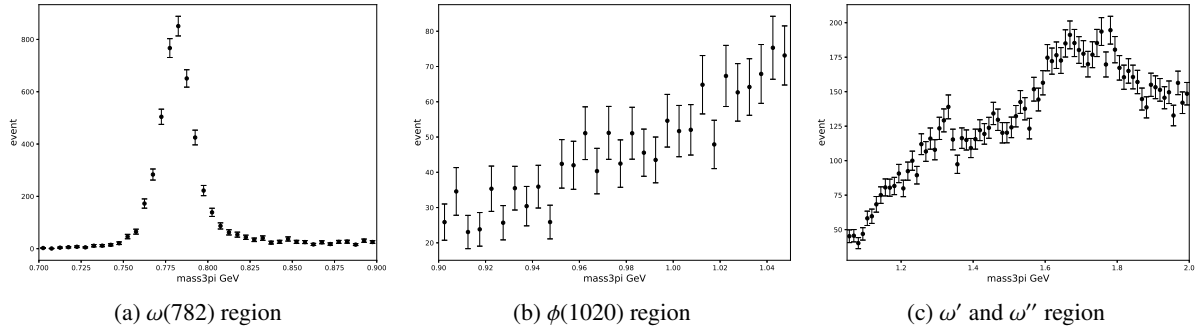


Figure 7: Estimated background contributions from  $e^+e^- \rightarrow \pi^+\pi^-\pi^0\pi^0$  in the signal region after scaling, using data from round 16. The three panels correspond to different  $3\pi$  invariant mass regions: (a) the  $\omega(782)$  region, (b) the  $\phi(1020)$  region, and (c) the higher-mass  $\omega'/\omega''$  region.



## 5.2 Estimate of Other Backgrounds and Signal Yields

The estimation of the dominant peaking backgrounds from  $e^+e^- \rightarrow \pi^+\pi^-\pi^0\pi^0$  has been completed in the previous subsection. These backgrounds exhibit shapes similar to the signal resonance peaks and are therefore categorized as peaking backgrounds.

In addition to the peaking component, a non-peaking continuum background component must also be accounted for in the analysis. The primary objective of this study is to measure the cross section for  $e^+e^- \rightarrow \pi^+\pi^-\pi^0$  via ISR. The ISR photon reduces the effective c.m. energy, allowing the measurement of the signal yield in narrow intervals of the  $3\pi$  invariant mass ( $M_{3\pi}$ ).

The  $M_{3\pi}$  spectrum is divided into three physically motivated regions, each subdivided into multiple bins for detailed study:

- The  $\omega(782)$  region (0.7–0.9 GeV/ $c^2$ ), divided into 40 bins.
- The  $\phi(1020)$  region (0.9–1.05 GeV/ $c^2$ ), divided into 30 bins.
- The  $\omega'/\omega''$  region (1.05–2.0 GeV/ $c^2$ ), divided into 76 bins.

For each small  $M_{3\pi}$  interval, the signal yield is extracted through an extended unbinned maximum-likelihood fit to the  $U_{\text{miss}}$  distribution, where  $U_{\text{miss}}$  is defined as the difference between the missing energy and the magnitude of the missing momentum ( $U_{\text{miss}} = E_{\text{miss}} - |\vec{p}_{\text{miss}}|$ ). The fit model incorporates probability density functions (PDFs) representing the signal and all background components.

To illustrate the composition of  $U_{\text{miss}}$  distributions across different  $M_{3\pi}$  regions, the inclusive MC simulation is examined. Figure 8 shows the  $U_{\text{miss}}$  distributions for the full  $\omega(782)$  region, the full  $\phi(1020)$  region, and the full  $\omega'/\omega''$  region respectively, with the peaking background contribution from  $e^+e^- \rightarrow \pi^+\pi^-\pi^0\pi^0$  stacked according to its expected yield. These distributions demonstrate the characteristic features of each mass region that motivate our fitting strategy.

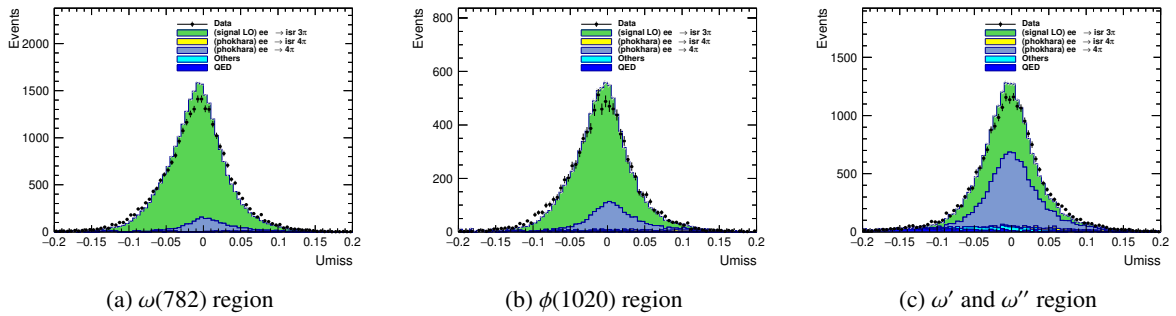


Figure 8:  $U_{\text{miss}}$  distributions from inclusive MC simulation for three  $M_{3\pi}$  regions: (a) the  $\omega(782)$  region, (b) the  $\phi(1020)$  region, and (c) the higher-mass  $\omega'/\omega''$  region.

Based on the understanding from the inclusive MC, the PDFs for the fit are constructed as follows:

- **Signal PDF:** Histogram shape from signal MC, convolved with a Gaussian for resolution correction.

- **Peaking background PDF:** Shape for  $e^+e^- \rightarrow \pi^+\pi^-\pi^0\pi^0$  from dedicated MC, convolved with the same Gaussian as signal. Yield constrained by Gaussian term using data-driven estimate from Section 5.
- **PDF shape sharing across bins:** Limited MC statistics preclude bin-by-bin shape extraction. Instead:
  - For the  $\omega(782)$  and  $\phi(1020)$  regions, where the  $M_{3\pi}$  bin width is  $5 \text{ MeV}/c^2$ , the PDF shapes are extracted from MC simulation every five consecutive bins, corresponding to  $25 \text{ MeV}/c^2$  intervals. All five bins within each such interval share the same PDF shape, under the assumption that the  $U_{\text{miss}}$  distribution varies slowly over this mass range.
  - For the  $\omega'/\omega''$  region, where the  $M_{3\pi}$  bin width is  $12.5 \text{ MeV}/c^2$ , the PDF shapes are also extracted every  $25 \text{ MeV}/c^2$ . Each extracted shape is therefore shared by two adjacent bins, balancing the need for shape precision against the limited MC statistics in this higher-mass region.
- **Continuum background PDF:** All remaining minor background processes are collectively described by a polynomial function. A first-order polynomial (linear function) is used for the  $\omega(782)$  and  $\phi(1020)$  regions, while a second-order Chebyshev polynomial is employed for the  $\omega'/\omega''$  region, as suggested by the shape of the inclusive MC distribution in Fig. 8c.

The fit procedure is validated by examining the quality of the fits to data in representative  $M_{3\pi}$  bins across the three mass regions. Figure 9 presents typical fit results for three selected bins, one from each region, demonstrating the ability of the fit model to accurately describe the data. The signal component, peaking background from  $\pi^+\pi^-\pi^0\pi^0$ , and polynomial continuum background are clearly distinguished, and the total PDF provides an excellent description of the data distribution in each case.

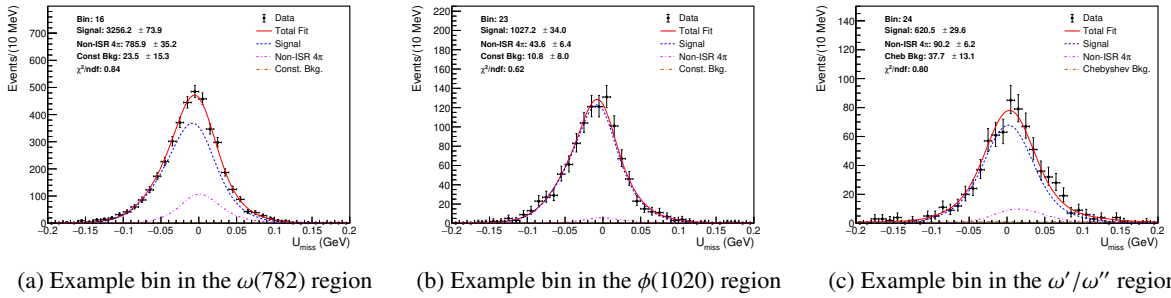


Figure 9: Typical  $U_{\text{miss}}$  fit results for representative  $M_{3\pi}$  bins from each of the three mass regions.

Applying this fitting procedure to all  $M_{3\pi}$  bins yields the extracted signal yields across the entire mass spectrum. Figure 10 shows the measured signal yields as a function of  $M_{3\pi}$  for the three regions. The  $\omega(782)$  resonance is clearly visible as a prominent peak around  $0.78 \text{ GeV}/c^2$ , while the  $\phi(1020)$  resonance appears as a distinct but smaller peak near  $1.02 \text{ GeV}/c^2$ . In the higher mass region, broad structures corresponding to the  $\omega'$  and  $\omega''$  excitations are observed, with significantly reduced yields due to the decreasing cross section and lower ISR photon probability at larger momentum transfers.

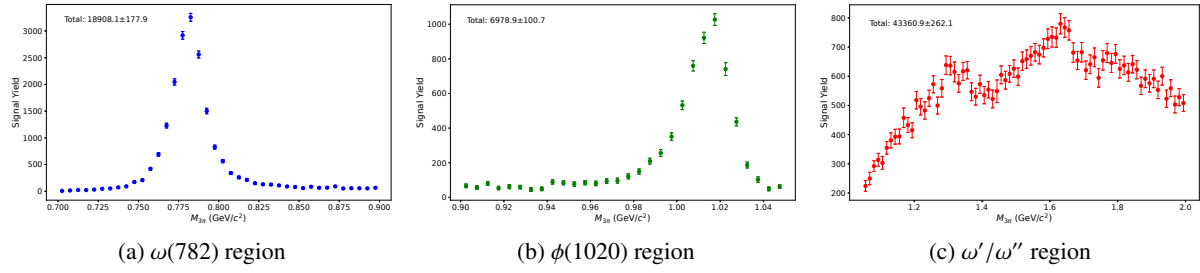


Figure 10: Extracted signal yields as a function of  $M_{3\pi}$  for the three mass regions. The error bars represent the statistical uncertainties from the fits. The  $\omega(782)$  and  $\phi(1020)$  resonances appear as clear peaks, while the higher-mass region reveals broader structures corresponding to excited  $\omega$  states.

The signal yields obtained from this fitting procedure provide the foundation for calculating the Born cross section for  $e^+e^- \rightarrow \pi^+\pi^-\pi^0$  as a function of  $M_{3\pi}$ . After applying efficiency corrections, unfolding for resolution effects, and subtracting residual background contributions, the cross section will be determined and compared with previous measurements and theoretical predictions.

## 6 Unfolding

To extract the dressed cross section  $\sigma_{3\pi}^{\text{dress}}(m)$ , we need the generator-level yield  $\left(\frac{dN}{dm}\right)^{\text{gen}}$ , which is related to the cross section within each mass interval by:

$$\left(\frac{dN}{dm}\right)^{\text{gen}} = \sigma_{3\pi}^{\text{dress}}(m) \cdot \frac{dL}{dm} \cdot \varepsilon \cdot \mathcal{B}(\pi^0 \rightarrow \gamma\gamma) \quad (6)$$

where  $\frac{dL}{dm}$  is the integrated luminosity in that interval, and  $\mathcal{B}(\pi^0 \rightarrow \gamma\gamma)$  is the branching fraction of the  $\pi^0 \rightarrow \gamma\gamma$  decay.

From data, we have obtained the observed signal yield  $\left(\frac{dN}{dm}\right)_i^{\text{obs}}$  in each mass interval by fitting the  $U_{\text{miss}}$  distribution, as shown in Fig. 10. Due to detector effects, events can migrate between different mass intervals, so these observed yields do not directly give the generator-level yields in Eq. (6). The migration is described by the response matrix  $P_{ij}$ , which relates the observed yield in bin  $i$  to the generator-level yield in bin  $j$ :

$$\left(\frac{dN}{dm}\right)_i^{\text{obs}} = \sum_j P_{ij} \left(\frac{dN}{dm}\right)_j^{\text{gen}} \quad (7)$$

Let  $A_{ij}$  denote the number of MC events that are generated in true bin  $j$  and reconstructed in bin  $i$ , where  $i$  and  $j$  index the observed and generator-level mass bins, respectively. The migration matrices  $A_{ij}$  for three representative mass regions are shown in Fig. 11, illustrating how events generated in a given true bin are distributed across observed bins due to detector effects. This matrix is then normalized column-wise to obtain the response matrix used in Eq. (7):

$$P_{ij} = \frac{A_{ij}}{\sum_{k=1}^{n_d} A_{kj}}, \quad \sum_{i=1}^{n_d} P_{ij} = 1, \quad (8)$$

where  $n_d$  is the number of reconstructed bins.

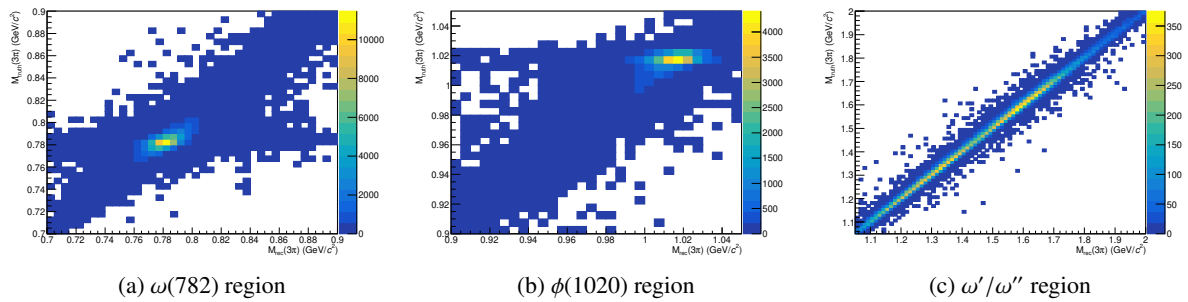


Figure 11: Response matrices  $A_{ij}$  for three representative mass regions, showing the migration of events between generator-level and reconstructed  $M_{3\pi}$  bins. The diagonal structure indicates good resolution, while off-diagonal entries reflect the effects of finite detector resolution and acceptance.

## 6.1 The IDS unfolding method

To extract the generator-level yield  $\left(\frac{dN}{dm}\right)_j^{\text{gen}}$  from the observed yields  $\left(\frac{dN}{dm}\right)_i^{\text{obs}}$  and the response matrix  $P_{ij}$ , we employ the Iterative Dynamically Stabilized (IDS) unfolding method [13]. This section introduces the mathematical formulation of the method; the specific implementation, parameter optimization, and iteration strategy are described in the following sections.

From the MC simulation, we have constructed the migration matrix  $A_{ij}$  (shown in Fig. 11) and the response matrix  $P_{ij}$  normalized column-wise as in Eq. (8). The corresponding unfolding matrix is defined row-wise:

$$\tilde{P}_{ij} = \frac{A_{ij}}{\sum_{k=1}^{n_u} A_{ik}}, \quad (9)$$

which gives the probability that an event reconstructed in bin  $i$  originated from true bin  $j$ .

The IDS method uses a regularization function  $f(|\Delta|, \tilde{\sigma}, \lambda)$  that dynamically distinguishes statistical fluctuations from genuine signals. Here,  $|\Delta|$  represents the absolute difference between data and the model prediction (e.g.,  $|\Delta d_i|$  for the discrepancy in reconstructed bin  $i$ , or  $|\Delta u_j|$  for the discrepancy in true bin  $j$ ),  $\tilde{\sigma}$  denotes the corresponding statistical uncertainty ( $\tilde{\sigma} d_i$  or  $\tilde{\sigma} u_j$ ), and  $\lambda$  is a tunable regularization parameter. The function satisfies the following asymptotic behavior:

$$\lim_{\frac{|\Delta|}{\lambda \tilde{\sigma}} \rightarrow 0} f(|\Delta|, \tilde{\sigma}, \lambda) = 0, \quad \lim_{\frac{|\Delta|}{\lambda \tilde{\sigma}} \rightarrow \infty} f(|\Delta|, \tilde{\sigma}, \lambda) = 1. \quad (10)$$

Thus, for discrepancies much smaller than  $\lambda$  times the uncertainty, the function approaches zero, treating the difference as a fluctuation; for discrepancies much larger, it approaches unity, indicating a real signal that should be unfolded. The transition between these regimes is smooth, avoiding sharp cutoffs that could introduce artifacts.

In the IDS method, the unfolding is performed iteratively. At iteration  $k$ , the unfolded spectrum  $u_j^{(k)}$  is given by:

$$u_j^{(k)} = \eta \cdot t_j + \sum_{i=1}^{n_d} \left[ f(|\Delta d_i^{(k-1)}|, \tilde{\sigma} d_i, \lambda_U) \Delta d_i^{(k-1)} \tilde{P}_{ij}^{(k-1)} + \left(1 - f(|\Delta d_i^{(k-1)}|, \tilde{\sigma} d_i, \lambda_U)\right) \Delta d_i^{(k-1)} \delta_{ij} \right], \quad (11)$$

where:

- $u_j^{(k)}$  is the unfolded generator-level yield in bin  $j$  at iteration  $k$ ,
- $t_j$  is the true MC yield in bin  $j$  from simulation,
- $\eta = N_d^{\text{MC}}/N_{\text{MC}}$  is the normalization factor that scales the MC to match the data in regions without significant new structures,
- $\Delta d_i^{(k-1)} = d_i - B_i^d - \eta \cdot r_i^{(k-1)}$  is the difference between the observed data yield  $d_i$  and the sum of the estimated background  $B_i^d$  (which may be zero if no background subtraction is applied) and the normalized reconstructed MC spectrum  $\eta \cdot r_i^{(k-1)}$ ,
- $r_i^{(k-1)}$  is the reconstructed MC spectrum at iteration  $k-1$ , obtained by folding the unfolded spectrum  $u_j^{(k-1)}$  with the response matrix  $P_{ij}^{(k-1)}$ ,

- $\tilde{\sigma}d_i$  is the statistical uncertainty of  $d_i$ ,
- $\tilde{P}_{ij}^{(k-1)}$  is the unfolding matrix from the previous iteration,
- $\lambda_U$  is the regularization parameter used in the unfolding step,
- $\delta_{ij}$  is the Kronecker delta, which equals 1 if  $i = j$  and 0 otherwise.

The first term on the right-hand side,  $\eta \cdot t_j$ , represents the normalized true MC spectrum, which serves as a baseline. The remaining sum dynamically partitions the discrepancy  $\Delta d_i^{(k-1)}$ : a fraction  $f$  is unfolded using  $\tilde{P}_{ij}^{(k-1)}$ , while the complementary fraction  $1 - f$  is retained in the original bin  $j = i$  via the Kronecker delta.

The response matrix itself can be iteratively improved. Using the difference  $\Delta u_j = u_j - \eta \cdot t_j$  between the current unfolded result and the normalized true MC, the migration matrix is modified as:

$$A'_{ij} = A_{ij} + f \left( |\Delta u_j|, \tilde{\sigma}u_j, \lambda_M \right) \Delta u_j P_{ij} \frac{N_{MC}}{N_d^{MC}}, \quad \text{for } i \in \{1, \dots, n_d\}, \quad (12)$$

where:

- $\Delta u_j$  is the difference between the current unfolded yield  $u_j$  and the normalized true MC  $\eta \cdot t_j$ ,
- $\tilde{\sigma}u_j$  is the statistical uncertainty associated with  $u_j$ ,
- $\lambda_M$  is the regularization parameter controlling the modification strength,
- $P_{ij}$  is the current response matrix (column-normalized  $A_{ij}$ ),
- $N_{MC} = \sum_j t_j$  is the total number of generated MC events,
- $N_d^{MC}$  is the number of data events compatible with the MC model (used for normalization).

The updated migration matrix  $A'_{ij}$  is then renormalized column-wise to obtain an improved response matrix  $P_{ij}$ , and row-wise to obtain an improved unfolding matrix  $\tilde{P}_{ij}$ .

## 6.2 Parameter optimization and figure of merit

The IDS method involves three regularization parameters:  $\lambda_L$  for the initial unfolding,  $\lambda_M$  for modifying the response matrix, and  $\lambda_U$  for subsequent unfoldings. The iterative procedure consists of the following three steps:

1. **Initial unfolding:** Perform an unfolding using Eq. (11) with  $\lambda = \lambda_L$  and the initial response matrix  $P_{ij}$ , obtaining  $u_j^{(0)}$ .
2. **Matrix improvement:** Modify the migration matrix via Eq. (12) using  $\lambda_M$  and the current  $u_j$ . Renormalize to obtain updated  $P_{ij}$  and  $\tilde{P}_{ij}$ .
3. **Improved unfolding:** Perform an unfolding using Eq. (11) with  $\lambda = \lambda_U$  and the updated matrices.

These three steps can be repeated for a chosen number of iterations.

In this analysis, the four parameters  $\lambda_M$ ,  $\lambda_L$ ,  $\lambda_U$ , and  $n_{\text{iter}}$  are optimized by scanning over a grid of possible values. For each parameter combination  $(\lambda_M, \lambda_L, \lambda_U, n_{\text{iter}})$ , we perform the IDS unfolding and evaluate the quality of the result using a  $\chi^2$  figure of merit. The  $\chi^2$  is computed by comparing the unfolded spectrum  $u_j$  to the true generator-level distribution  $t_j^{\text{truth}}$  from MC simulation. The MC sample used for this comparison has been normalized such that its total integral matches the total number of signal events observed in data (shown in Fig. 10), ensuring that the statistical uncertainties from the toy ensembles properly reflect the data statistics.

$$\chi^2 = (\mathbf{u} - \mathbf{t}^{\text{truth}})^T \Sigma^{-1} (\mathbf{u} - \mathbf{t}^{\text{truth}}), \quad (13)$$

where  $\mathbf{u}$  and  $\mathbf{t}^{\text{truth}}$  are vectors of bin contents, and  $\Sigma$  is the covariance matrix of the unfolded spectrum. The covariance matrix accounts for the statistical uncertainties and their correlations introduced by the unfolding procedure. It is estimated using a MC sampling method:

1. Generate an ensemble of  $N_{\text{toy}}$  pseudo-data distributions by fluctuating the original observed spectrum according to its statistical uncertainties, using a random number generator.
2. For each pseudo-data distribution, perform the IDS unfolding with the same parameter combination  $(\lambda_L, \lambda_U, n_{\text{iter}})$ , obtaining an unfolded spectrum  $u_j^{(k)}$  for toy  $k$ .
3. From the ensemble of unfolded spectra, compute the covariance matrix:

$$\Sigma_{ij} = \frac{1}{N_{\text{toy}} - 1} \sum_{k=1}^{N_{\text{toy}}} (u_i^{(k)} - \bar{u}_i) (u_j^{(k)} - \bar{u}_j), \quad (14)$$

where  $\bar{u}_i = \frac{1}{N_{\text{toy}}} \sum_k u_i^{(k)}$  is the mean unfolded yield in bin  $i$  across the toy ensemble.

In cases where the covariance matrix is singular or ill-conditioned, a small diagonal perturbation  $\lambda_{\text{reg}} I$  is added before inversion to ensure numerical stability, with  $\lambda_{\text{reg}} = 10^{-6}$  in this analysis.

The optimal parameter combination is chosen based on several considerations: the number of iterations  $n_{\text{iter}}$  should be as small as possible to avoid unnecessary complexity; the  $\chi^2$  value defined in Eq. (13) should be small, indicating good agreement between the unfolded and true spectra; and the  $\chi^2$  per degree of freedom,  $\chi^2/n_{\text{dof}}$ , should be close to or less than one, indicating that the uncertainties are well estimated. Using the MC sample as a stand-in for data, a scan is performed over a grid of  $\lambda_L$ ,  $\lambda_U$ , and  $n_{\text{iter}}$  values, and the combination that best satisfies these criteria is selected for the final unfolding of the data. The optimized parameters obtained from this procedure for the three representative mass regions are summarized in Table 4. The unfolding results obtained with these optimized parameters are shown in Fig. 12.

To validate the unfolding procedure, we compare the relative statistical uncertainty of the unfolded spectrum with that of the test MC sample. The relative uncertainty of the test MC spectrum,  $\delta_{\text{test}}$ , is computed by summing in quadrature the Poisson errors of each bin and dividing by the total yield:

$$\delta_{\text{test}} = \frac{\sqrt{\sum_i \sigma_i^2}}{\sum_i N_i}, \quad (15)$$

Table 4: Optimized unfolding parameters for the three representative mass regions. For each region, the parameters  $\lambda_L$ ,  $\lambda_U$ ,  $\lambda_M$ , and number of iterations  $n_{\text{iter}}$  were selected based on the  $\chi^2$  minimization procedure described in the text. The resulting  $\chi^2/\text{ndf}$  values indicate good agreement between the unfolded and true MC distributions.

| Region             | $\lambda_L$        | $\lambda_U$        | $\lambda_M$        | $n_{\text{iter}}$ | $\chi^2/\text{ndf}$ |
|--------------------|--------------------|--------------------|--------------------|-------------------|---------------------|
| $\omega(782)$      | $1 \times 10^{-3}$ | $1 \times 10^{-4}$ | 1                  | 4                 | 28.31/40            |
| $\phi(1020)$       | $1 \times 10^{-4}$ | $1 \times 10^{-4}$ | $1 \times 10^{-3}$ | 4                 | 23.86/30            |
| $\omega'/\omega''$ | $1 \times 10^{-3}$ | 1                  | 11                 | 5                 | 76.18/76            |

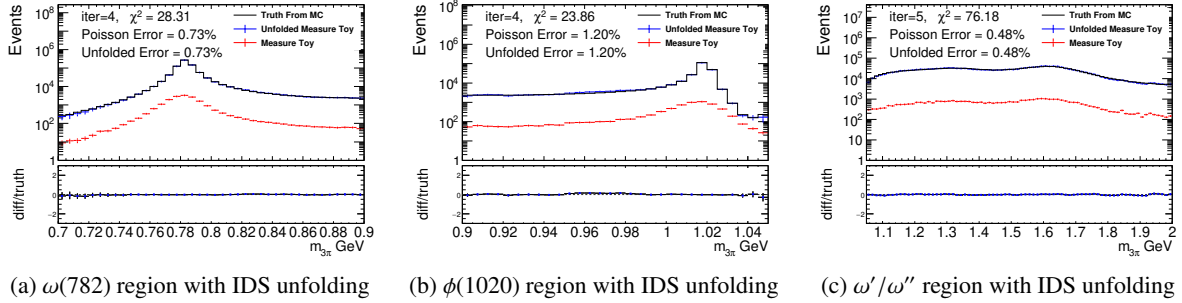


Figure 12: Unfolded results for three representative mass regions using the IDS method with optimized parameters. Each panel shows the true MC distribution (black), the measured MC distribution (red), and the unfolded result (blue points) with its statistical uncertainties. The MC sample has been normalized to match the data statistics shown in Fig. 10, ensuring that the uncertainties properly reflect the data statistical power.

where  $N_i$  and  $\sigma_i$  are the content and Poisson error of bin  $i$  in the test MC distribution, respectively. This represents the Poisson statistical uncertainty of the test MC sample.

The relative uncertainty of the unfolded spectrum,  $\delta_{\text{unfolded}}$ , is derived from the full covariance matrix  $\Sigma$  obtained from the toy MC ensemble:

$$\delta_{\text{unfolded}} = \frac{\sqrt{\sum_{i,j} \Sigma_{ij}}}{\sum_i u_i}, \quad (16)$$

where  $u_i$  are the bin contents of the unfolded spectrum. The relative uncertainty of the unfolded spectrum should not be smaller than the Poisson uncertainty of the test MC sample.



## 7 Correction of response matrix, and unfolding of data

With the optimized unfolding parameters determined from the MC validation studies (summarized in Table 4), we now proceed to unfold the data. However, before applying the IDS method to data, several corrections must be applied to the response matrix derived from MC simulation to account for discrepancies between data and MC. These discrepancies arise from various sources: the tracking efficiency of charged particles, PID performance, the reconstruction efficiency of neutral  $\pi^0$  mesons, the detection efficiency of ISR photons, and differences in the production rate of additional ISR photons (NLO) between the Phokhara generator and data. The following subsections address each of these corrections in detail.

### 7.1 Tracking and PID efficiencies

The tracking and PID efficiencies for charged pions are adopted from the dedicated study in Ref. [14], where they are measured using  $D^0\bar{D}^0$  and  $D^+D^-$  events. The kinematic coverage of that study matches the momentum range of charged pions in this analysis. For each momentum bin, the correction factors  $c_{\text{trk}}(p)$  and  $c_{\text{pid}}(p)$  are derived separately for tracking and PID.

### 7.2 The $\pi^0$ reconstruction efficiency

The reconstruction efficiency for  $\pi^0$  mesons is studied using a dedicated control sample of  $e^+e^- \rightarrow \omega\pi_{\text{tag}}^0$  events at  $\sqrt{s} = 3.773$  GeV, with  $\omega \rightarrow \pi^+\pi^-\pi_{\text{pred}}^0$ . This process provides a clean environment to study  $\pi^0$  efficiency due to the distinct kinematic properties of the two neutral pions: the  $\pi_{\text{tag}}^0$  carries approximately half of the center-of-mass momentum, while the  $\pi_{\text{pred}}^0$  from  $\omega$  decay has a momentum spectrum ranging from 0 to  $\sqrt{s}/2$ , making it well suited for a momentum-dependent efficiency study.

The measurement is performed independently for each data-taking period (rounds 0304, 15, 16, and 17) using the 3.773 GeV data sample. The selection criteria are as follows:

- The selection requirements for charged pions and  $\pi^0$  reconstruction are identical to those used in the signal channel  $e^+e^- \rightarrow \pi^+\pi^-\pi^0$ .
- Events are required to have at least one reconstructed  $\pi^0$  candidate.
- In events with multiple  $\pi^0$  candidates, the two with the best fit quality from 1C kinematic fitting are retained for further analysis.
- The  $\pi_{\text{tag}}^0$  is identified as the  $\pi^0$  candidate whose energy is closest to half the center-of-mass energy ( $\sqrt{s}/2$ ), with a requirement that the difference be within 0.3 GeV.
- A 1C kinematic fit is then performed: the recoil momentum of the  $\pi^+\pi^-\pi_{\text{tag}}^0$  system against the initial  $e^+e^-$  system is calculated in the center-of-mass frame; this recoil momentum is constrained to the  $\pi^0$  mass, corresponding to the missing  $\pi_{\text{pred}}^0$ ; subsequently, the  $\pi^+\pi^-$  system together with the  $\pi_{\text{tag}}^0$  is constrained to the center-of-mass system.

The 1C kinematic fit is required to satisfy  $\chi^2 < 15$ . After the fit, the four-momentum of the  $\pi_{\text{pred}}^0$  is updated. The invariant mass of the  $\pi^+\pi^-\pi_{\text{pred}}^0$  system is then calculated, and events are required to have this invariant mass within the  $\omega$  meson mass window of 0.7–0.86 GeV.

Before applying the momentum-dependent efficiency study, we first examine the overall  $\pi^+\pi^-\pi_{\text{pred}}^0$  invariant mass distribution without requiring a matching reconstructed  $\pi^0$  in the predicted direction. Figure 13a shows this distribution for all events passing the selection criteria, with data, MC, and estimated background contributions overlaid. The signal region 0.7–0.86 GeV exhibits a clean peak with negligible background, confirming the purity of the control sample.

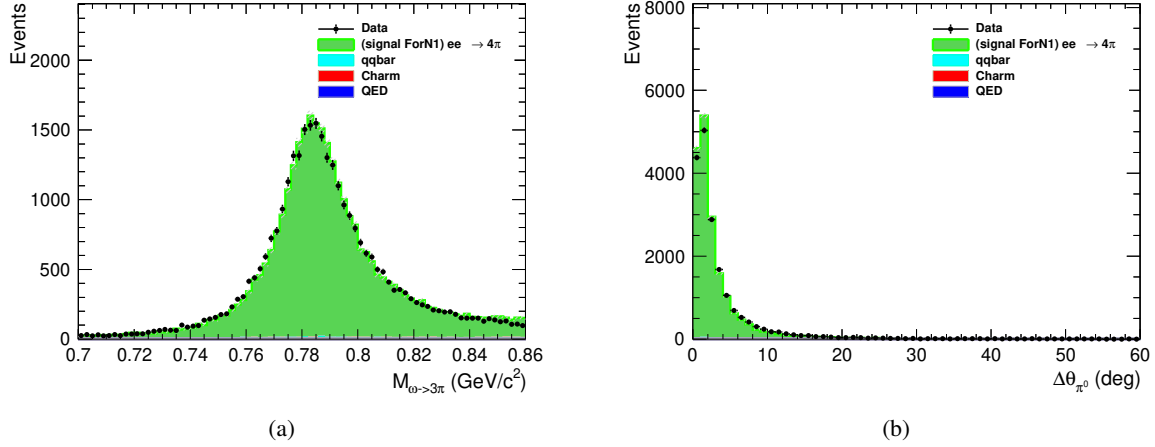


Figure 13: (a) Full  $\pi^+\pi^-\pi_{\text{pred}}^0$  invariant mass distribution for all selected events, showing data (points), MC (green), and estimated background (other colors). The vertical dashed lines indicate the  $\omega$  mass window used for event selection. (b) Angular separation  $\Delta\theta$  between the predicted  $\pi_{\text{pred}}^0$  direction and additional reconstructed  $\pi^0$  candidates. The vertical dashed line indicates the  $20^\circ$  matching cone used in this analysis.

To associate a reconstructed  $\pi^0$  with the predicted  $\pi_{\text{pred}}^0$ , we examine the angular separation  $\Delta\theta$  between the direction of the predicted  $\pi_{\text{pred}}^0$  and any additional reconstructed  $\pi^0$  candidate (excluding the  $\pi_{\text{tag}}^0$ ). Figure 13b shows the distribution of this angular separation for events where an additional  $\pi^0$  is found. A clear peak at small angles is observed, corresponding to correctly matched  $\pi^0$  candidates. Based on this distribution, we define a matching cone of  $20^\circ$ : if an additional reconstructed  $\pi^0$  lies within  $20^\circ$  of the predicted direction, it is considered as the reconstructed  $\pi_{\text{pred}}^0$ .

For each momentum bin, Fig. 14 shows the  $\pi^+\pi^-\pi_{\text{pred}}^0$  invariant mass distribution for events where a matching reconstructed  $\pi^0$  is found within the  $20^\circ$  cone. The distributions for  $p_{\pi^0} < 0.2 \text{ GeV}$ ,  $0.2 \leq p_{\pi^0} < 0.4 \text{ GeV}$ ,  $0.4 \leq p_{\pi^0} < 0.6 \text{ GeV}$ ,  $0.6 \leq p_{\pi^0} < 0.8 \text{ GeV}$ ,  $0.8 \leq p_{\pi^0} < 1.0 \text{ GeV}$ ,  $1.0 \leq p_{\pi^0} < 1.2 \text{ GeV}$ ,  $1.2 \leq p_{\pi^0} < 1.4 \text{ GeV}$ , and  $p_{\pi^0} \geq 1.4 \text{ GeV}$  all show clean signal peaks, demonstrating that the matching procedure is effective for the entire momentum range.

Given the negligible background in the signal region, the  $\pi^0$  reconstruction efficiency can be directly determined by counting events. For each momentum bin, we define:

- $N_{\text{data}}^{\text{total}}(p)$ : number of data events in the  $\omega$  mass window, regardless of whether a matching  $\pi^0$  is found.
- $N_{\text{data}}^{\text{matched}}(p)$ : number of data events in the  $\omega$  mass window with a matching reconstructed  $\pi^0$  within  $20^\circ$  of the predicted direction.

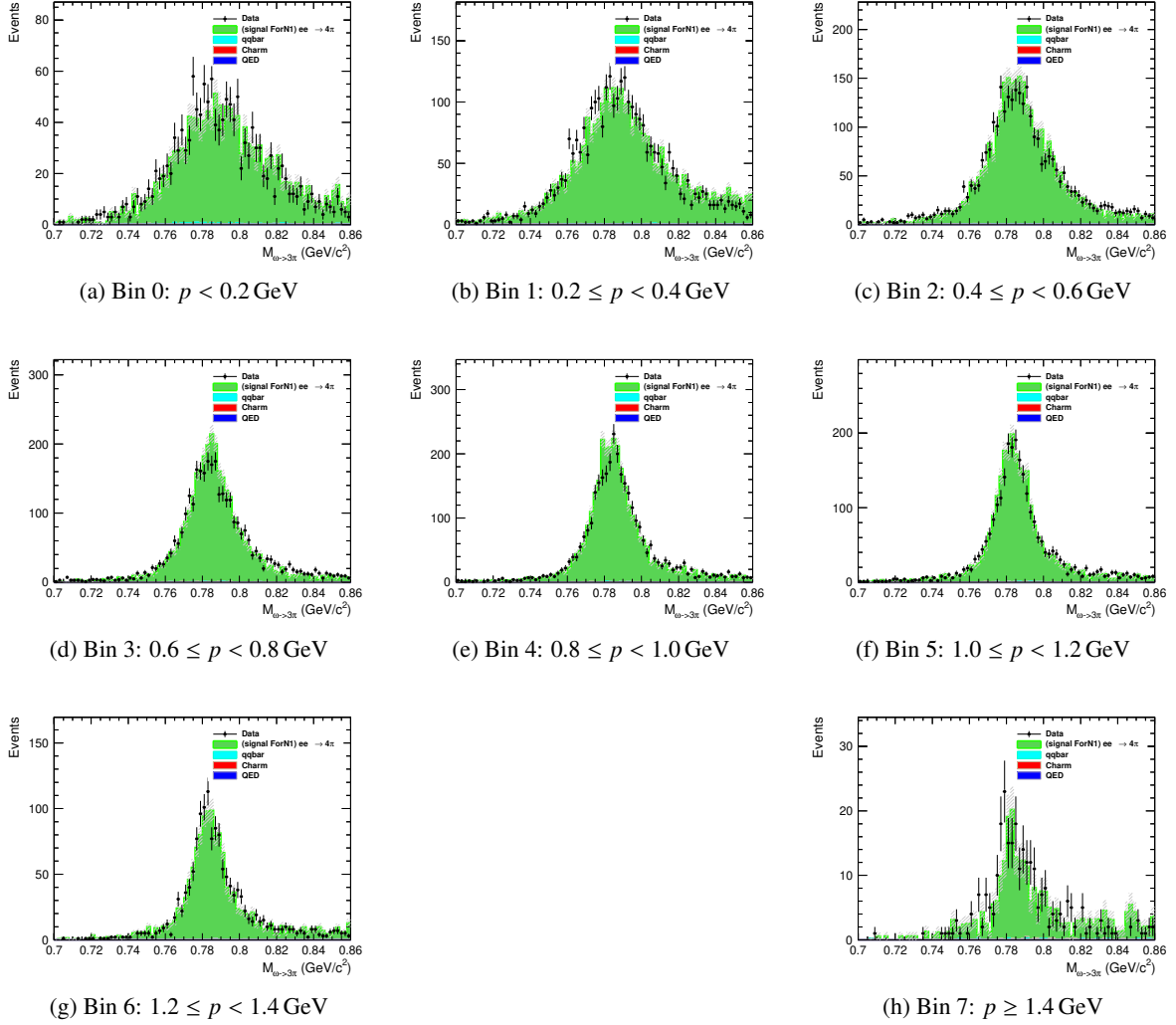


Figure 14:  $\pi^+\pi^-\pi^0_{\text{pred}}$  invariant mass distributions for events with a matching reconstructed  $\pi^0$  within  $20^\circ$ , shown for each  $\pi^0_{\text{pred}}$  momentum bin. Data (points), MC (green), and estimated background (other colors) are overlaid. The vertical dashed lines indicate the  $\omega$  mass window used for event selection.

- $N_{\text{MC}}^{\text{total}}(p)$  and  $N_{\text{MC}}^{\text{matched}}(p)$ : corresponding quantities for MC.

The efficiencies in data and MC are then computed as:

$$\varepsilon_{\text{data}}(p) = \frac{N_{\text{data}}^{\text{matched}}(p)}{N_{\text{data}}^{\text{total}}(p)}, \quad \varepsilon_{\text{MC}}(p) = \frac{N_{\text{MC}}^{\text{matched}}(p)}{N_{\text{MC}}^{\text{total}}(p)}. \quad (17)$$

The correction factor for  $\pi^0$  reconstruction is then derived as the ratio of data efficiency to MC efficiency:

$$c_{\pi^0}(p) = \frac{\varepsilon_{\text{data}}(p)}{\varepsilon_{\text{MC}}(p)}. \quad (18)$$

The calculated efficiencies and correction factors are presented in Table 5. Each entry is shown as central value  $\pm$  its statistical uncertainty.

Table 5: The  $\pi^0$  reconstruction efficiencies in data and MC, and the derived correction factors with their statistical uncertainties for different  $\pi^0$  momentum intervals.

| $p_{\pi^0}$ (GeV) | $\varepsilon_{\text{data}}$ | $\varepsilon_{\text{MC}}$ | $c_{\pi^0}$         |
|-------------------|-----------------------------|---------------------------|---------------------|
| $< 0.2$           | $0.3892 \pm 0.0080$         | $0.3958 \pm 0.0064$       | $0.9833 \pm 0.0254$ |
| $0.2 - 0.4$       | $0.4579 \pm 0.0061$         | $0.4782 \pm 0.0050$       | $0.9575 \pm 0.0168$ |
| $0.4 - 0.6$       | $0.5442 \pm 0.0067$         | $0.5636 \pm 0.0059$       | $0.9656 \pm 0.0163$ |
| $0.6 - 0.8$       | $0.6265 \pm 0.0068$         | $0.6471 \pm 0.0060$       | $0.9682 \pm 0.0150$ |
| $0.8 - 1.0$       | $0.6946 \pm 0.0068$         | $0.7124 \pm 0.0060$       | $0.9750 \pm 0.0142$ |
| $1.0 - 1.2$       | $0.7540 \pm 0.0074$         | $0.7785 \pm 0.0064$       | $0.9685 \pm 0.0139$ |
| $1.2 - 1.4$       | $0.8089 \pm 0.0094$         | $0.8161 \pm 0.0082$       | $0.9912 \pm 0.0167$ |
| $\geq 1.4$        | $0.8017 \pm 0.0215$         | $0.8574 \pm 0.0155$       | $0.9350 \pm 0.0283$ |

### 7.3 The ISR photon detection efficiency

The ISR photon detection efficiency is studied using the same control sample of  $e^+e^- \rightarrow \omega\pi^0$  events at  $\sqrt{s} = 3.773$  GeV, with  $\omega \rightarrow \pi^+\pi^-\pi^0$ . Unlike the previous  $\pi^0$  efficiency study which tagged one  $\pi^0$  and studied the other from  $\omega$  decay, here we identify one  $\pi^0$  as coming from  $\omega$  decay and study the photon detection efficiency using the two photons from the decay of the remaining  $\pi^0$ . This remaining  $\pi^0$ , which balances the  $\omega$  in the event, carries approximately half of the center-of-mass energy ( $\sqrt{s}/2$ ).

The measurement is performed independently for each data-taking period using the 3.773 GeV data sample. The selection criteria are as follows:

- The selection requirements for charged pions are identical to those used in the signal channel  $e^+e^- \rightarrow \pi^+\pi^-\pi^0$ .
- Events are required to have at least one reconstructed  $\pi^0$  candidate.

- For each reconstructed  $\pi^0$  candidate, the invariant mass of the  $\pi^+\pi^-$  system combined with this  $\pi^0$  is calculated. Among all  $\pi^0$  candidates, the one that yields an invariant mass closest to the nominal  $\omega$  mass and within  $\pm 0.3$  GeV is identified as originating from the  $\omega$  decay ( $\pi_\omega^0$ ).
- The other  $\pi^0$  in the event (denoted as  $\pi_{\text{bal}}^0$ ) balances the  $\omega$  system and has a characteristic energy of approximately  $\sqrt{s}/2$ . It decays into two photons:  $\pi_{\text{bal}}^0 \rightarrow \gamma\gamma$ . Note that this  $\pi_{\text{bal}}^0$  itself is not required to be reconstructed; we only study its decay photons, and its existence is inferred from the  $\omega$  system and momentum conservation.
- All remaining photon candidates in the event (excluding those used in  $\pi_\omega^0$  reconstruction) are considered as potential tagged photons. For each such photon candidate, we treat it as the tagged photon ( $\gamma_{\text{tag}}$ ) and perform a 2C kinematic fit where the other photon ( $\gamma_{\text{pred}}$ ) is inferred via momentum conservation.
- A 2C kinematic fit is performed for each photon candidate with the following constraints:
  - The invariant mass of the missing momentum (corresponding to  $\gamma_{\text{pred}}$ ) is constrained to zero (photon mass).
  - The invariant mass of the tagged photon and the predicted photon is constrained to the  $\pi^0$  mass.
  - The total four-momentum of the event is constrained to the center-of-mass system ( $\sqrt{s} = 3.773$  GeV).
- Entries with  $\chi^2 < 5$  from the kinematic fit are classified as good entries. Events are required to have at least one good entry.
- For events with multiple good entries, the two with the smallest  $\chi^2$  are retained. From these two, the entry with the smaller tagged photon energy (correspondingly larger predicted photon energy) is selected for further analysis. This selection ensures that the predicted photon energy lies in the region of interest above 1 GeV.
- The selected entry is required to satisfy the kinematic fit quality  $\chi^2 < 5$ .

Before applying the energy-dependent efficiency study, we first examine the overall  $\pi^+\pi^-\pi_\omega^0$  invariant mass distribution without requiring a matching reconstructed photon in the predicted direction. Figure 15a shows this distribution for all events passing the selection criteria, with data, MC, and estimated background contributions overlaid. The signal region 0.7–0.86 GeV exhibits a clean peak with negligible background, confirming the purity of the control sample.

To associate a reconstructed photon with the predicted photon  $\gamma_{\text{pred}}$ , we examine the angular separation  $\Delta\theta$  between the direction of the predicted photon and any reconstructed photon candidate (excluding those used in  $\pi_\omega^0$  reconstruction). Figure 15b shows the distribution of this angular separation for events where a photon is found. A clear peak at small angles is observed, corresponding to correctly matched photon candidates. Based on this distribution, we define a matching cone of  $4^\circ$ : if a reconstructed photon lies within  $4^\circ$  of the predicted direction, it is considered as the reconstructed  $\gamma_{\text{pred}}$ .

For each energy bin, Fig. 16 shows the  $\pi^+\pi^-\pi_\omega^0$  invariant mass distribution for events where a matching reconstructed photon is found within the  $4^\circ$  cone. The distributions for  $1.0 \leq E_\gamma < 1.1$  GeV,

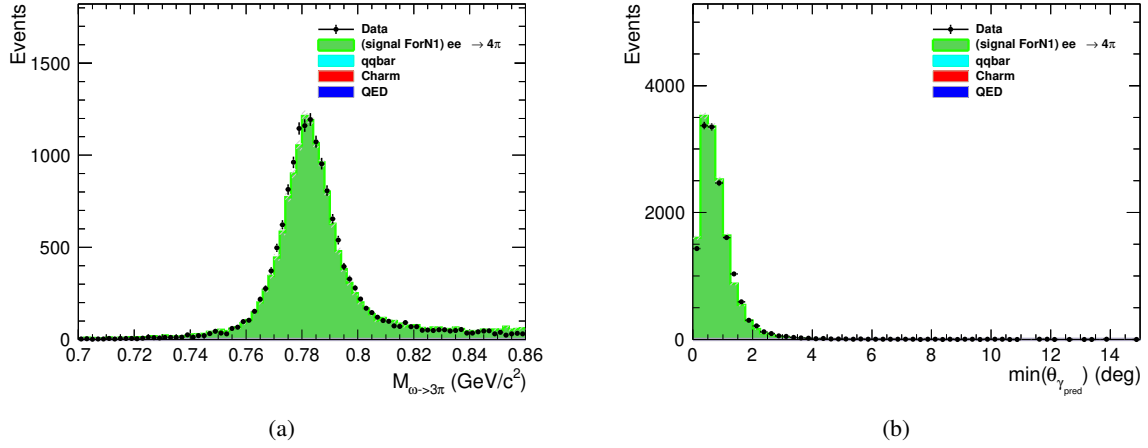


Figure 15: (a) Full  $\pi^+\pi^-\pi_\omega^0$  invariant mass distribution for all selected events, showing data (points), MC (green), and estimated background (other colors). The vertical dashed lines indicate the  $\omega$  mass window used for event selection. (b) Angular separation  $\Delta\theta$  between the predicted photon  $\gamma_{\text{pred}}$  direction and reconstructed photon candidates. The vertical dashed line indicates the  $4^\circ$  matching cone used in this analysis.

1.1  $\leq E_\gamma < 1.2$  GeV, 1.2  $\leq E_\gamma < 1.3$  GeV, 1.3  $\leq E_\gamma < 1.4$  GeV, 1.4  $\leq E_\gamma < 1.5$  GeV, 1.5  $\leq E_\gamma < 1.6$  GeV, 1.6  $\leq E_\gamma < 1.7$  GeV, and  $E_\gamma \geq 1.8$  GeV all show clean signal peaks, demonstrating that the matching procedure is effective for the entire energy range.

Given the negligible background in the signal region, the photon detection efficiency can be directly determined by counting events. For each energy bin, we define:

- $N_{\text{data}}^{\text{total}}(E)$ : number of data events in the  $\omega$  mass window, regardless of whether a matching photon is found.
- $N_{\text{data}}^{\text{matched}}(E)$ : number of data events in the  $\omega$  mass window with a matching reconstructed photon within  $4^\circ$  of the predicted direction.
- $N_{\text{MC}}^{\text{total}}(E)$  and  $N_{\text{MC}}^{\text{matched}}(E)$ : corresponding quantities for MC.

The efficiencies in data and MC are then computed as:

$$\varepsilon_{\text{data}}(E) = \frac{N_{\text{data}}^{\text{matched}}(E)}{N_{\text{data}}^{\text{total}}(E)}, \quad \varepsilon_{\text{MC}}(E) = \frac{N_{\text{MC}}^{\text{matched}}(E)}{N_{\text{MC}}^{\text{total}}(E)}. \quad (19)$$

The correction factor for photon detection is then derived as the ratio of data efficiency to MC efficiency:

$$c_\gamma(E) = \frac{\varepsilon_{\text{data}}(E)}{\varepsilon_{\text{MC}}(E)}. \quad (20)$$

The calculated efficiencies and correction factors are presented in Table 6. Each entry is shown as central value  $\pm$  its statistical uncertainty.

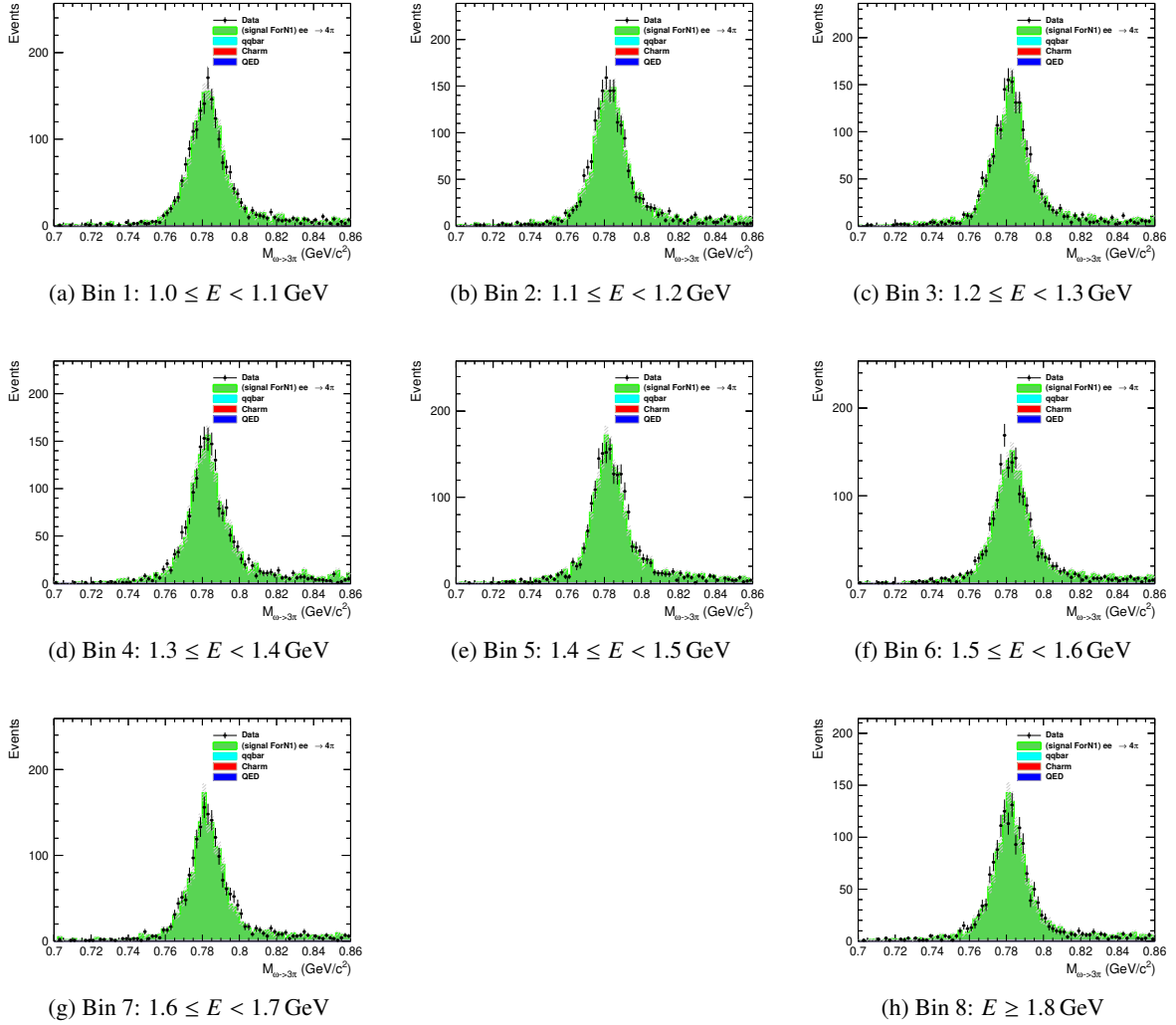


Figure 16:  $\pi^+\pi^-\pi^0$  invariant mass distributions for events with a matching reconstructed photon within  $4^\circ$ , shown for each predicted photon energy bin. Data (points), MC (green), and estimated background (other colors) are overlaid. The vertical dashed lines indicate the  $\omega$  mass window used for event selection.

Table 6: The photon detection efficiencies in data and MC, and the derived correction factors with their statistical uncertainties for different photon energy intervals. The correction factor uncertainties are calculated using error propagation.

| $E_\gamma$ (GeV) | $\varepsilon_{\text{data}}$ | $\varepsilon_{\text{MC}}$ | $c_\gamma$          |
|------------------|-----------------------------|---------------------------|---------------------|
| 1.0 – 1.1        | $0.9744 \pm 0.0036$         | $0.9791 \pm 0.0027$       | $0.9952 \pm 0.0045$ |
| 1.1 – 1.2        | $0.9726 \pm 0.0038$         | $0.9799 \pm 0.0026$       | $0.9926 \pm 0.0046$ |
| 1.2 – 1.3        | $0.9755 \pm 0.0035$         | $0.9798 \pm 0.0026$       | $0.9956 \pm 0.0044$ |
| 1.3 – 1.4        | $0.9724 \pm 0.0037$         | $0.9799 \pm 0.0026$       | $0.9923 \pm 0.0045$ |
| 1.4 – 1.5        | $0.9748 \pm 0.0035$         | $0.9821 \pm 0.0025$       | $0.9926 \pm 0.0043$ |
| 1.5 – 1.6        | $0.9796 \pm 0.0032$         | $0.9873 \pm 0.0021$       | $0.9922 \pm 0.0039$ |
| 1.6 – 1.7        | $0.9690 \pm 0.0040$         | $0.9846 \pm 0.0024$       | $0.9842 \pm 0.0047$ |
| $\geq 1.8$       | $0.9740 \pm 0.0040$         | $0.9824 \pm 0.0028$       | $0.9915 \pm 0.0049$ |

## 7.4 PHOKHARA Extra Photon Fraction

A BaBar study of the PHOKHARA generator used for signal simulation found a 20% excess of events with an additional energetic ISR photon along the beam line, in PHOKHARA relative to data in their  $e^+e^- \rightarrow \pi^+\pi^-$  process [15]. The ISR-based analysis at BaBar is not impacted as it relied on a different generator, but given that our signal process  $e^+e^- \rightarrow \pi^+\pi^-\pi^0$  is also simulated using PHOKHARA and involves ISR photon emission, this discrepancy warrants a cross-check in our analysis.

To investigate this effect in our channel, we examine the fraction of events with an additional ISR photon directly within our signal process  $e^+e^- \rightarrow \pi^+\pi^-\pi^0\gamma_{\text{ISR}}$ . This corresponds to studying events of the type  $e^+e^- \rightarrow \pi^+\pi^-\pi^0\gamma_{\text{ISR}}\gamma_{\text{NLO}}$ , where two ISR photons are present in the final state. NNLO contributions are not considered, as their production probability is significantly lower and the additional photons are typically too soft to be reliably identified given our detector resolution and event statistics. Moreover, the limited data sample size does not provide sufficient statistical power to meaningfully extract an NNLO contribution.

In this study, we rely on the ISR photon identified through the 4C kinematic fit procedure described in Sec. 4, which selects the optimal candidate event and tags the leading order (LO) ISR photon with high confidence. Based on this tagged photon, we proceed to investigate the presence of additional photons in the event.

Due to the finite coverage of the BESIII electromagnetic calorimeter, the study of NLO photons is divided into two independent categories based on their angular distribution:

- **Large Angle (LA) category:** NLO photons emitted at large angles fall within the detector acceptance and can be directly reconstructed from electromagnetic showers. In the candidate selection, no special treatment is applied beyond the standard reconstruction: among all photon candidates in the event, the one that minimizes the 4C kinematic fit  $\chi^2$  when included in the fit is identified as the NLO LA photon candidate. The kinematic fit for LA events is performed in the laboratory



frame, constraining the total four-momentum of all final state particles ( $\pi^+$ ,  $\pi^-$ ,  $\pi^0$ , the LO ISR photon, and the NLO LA photon) to the initial  $e^+e^-$  center-of-mass energy.

- **Small Angle (SA) category:** NLO photons emitted at very small angles along the beam direction escape detection, and their kinematic properties must be inferred indirectly. The momentum of this undetected photon is calculated from the missing momentum, with the constraint that its transverse components are zero in the center-of-mass frame ( $p_x^{\text{CM}} = p_y^{\text{CM}} = 0$ ). It is important to note that the beam direction in the laboratory frame is not aligned with the definition of transverse momentum in this context; therefore, we first boost the event to the center-of-mass frame, enforce the transverse momentum condition, and then boost back to the laboratory frame to obtain the four-momentum of the SA photon. The particle is further constrained to have zero mass (photon identity). A 3C kinematic fit is then performed in the laboratory frame, including the detected particles  $\pi^+$ ,  $\pi^-$ ,  $\pi^0$ , and the LO ISR photon, together with the inferred SA photon, to enforce overall energy and momentum conservation.

#### 7.4.1 Results for Large Angle (LA) NLO photons

#### 7.4.2 Results for Small Angle (SA) NLO photons

to fill

555 **8 Systematic uncertainties**

556 to fill

557 **9 Summary**

558 to fill

## References

- [1] M. Ablikim *et al.* (BESIII Collaboration), *Future Physics Programme of BESIII*, *Chinese Phys. C* **44** (2020).
- [2] R. X. Yang *et al.*, *MRPC detector for the BESIII E-TOF upgrade*, *Journal of Instrumentation*, **Volume 9**, September 2014. DOI: [10.1088/1748-0221/9/09/C09032](https://doi.org/10.1088/1748-0221/9/09/C09032).
- [3] M. Ablikim *et al.* [BESIII], DOI: [10.1016/j.nima.2009.12.050](https://doi.org/10.1016/j.nima.2009.12.050).
- [4] M. Ablikim *et al.* [BESIII], DOI: [10.1016/j.physletb.2015.11.043](https://doi.org/10.1016/j.physletb.2015.11.043).
- [5] M. Ablikim *et al.* [BESIII Collaboration], DOI: [10.1088/1674-1137/ad70a0](https://doi.org/10.1088/1674-1137/ad70a0).
- [6] S. Agostinelli *et al.* [GEANT4], *Nucl. Instrum. Meth. A* **506**, 250-303 (2003).
- [7] BESIII Collaboration, DocDB: 1483.
- [8] H. Czyż, A. Grzełińska, J. H. Kühn, and G. Rodrigo, DOI: [10.1140/epjc/s2006-02614-7](https://doi.org/10.1140/epjc/s2006-02614-7).
- [9] G. Rodrigo, A. Gehrmann-De Ridder, M. Guillaume, and J. H. Kühn, DOI: [10.1007/s100520100784](https://doi.org/10.1007/s100520100784).
- [10] G. Rodrigo, H. Czyż, J. H. Kühn, and M. Szopa, DOI: [10.1007/s100520200912](https://doi.org/10.1007/s100520200912).
- [11] E. A. Kuraev and V. S. Fadin, INSPIRE: 217313.
- [12] J. P. Lees *et al.* (BABAR Collaboration), DOI: [10.1103/PhysRevD.96.092009](https://doi.org/10.1103/PhysRevD.96.092009).
- [13] B. Malaescu, “An iterative, dynamically stabilized method of data unfolding,” [arXiv:0907.3791](https://arxiv.org/abs/0907.3791) [physics.data-an] (2009).
- [14] BESIII Collaboration, DocDB: 1497
- [15] J. P. Lees *et al.* (BABAR Collaboration), *Phys. Rev. D* **108**, L111103 (2023). DOI: [10.1103/PhysRevD.108.L111103](https://doi.org/10.1103/PhysRevD.108.L111103).

## APPENDIX